

Cover Letter

Date: June 16, 2026

To: Department of Computer Science, Namal University

From: Junaid Ameer Khan Roll No: NUM-BSCS-2022-46 BS Computer Science Program, 6th and 8th Semester, Spring 2026

Subject: Submission of Complex Computing Problem (CCP) Solution

Dear Instructor,

Please find enclosed the comprehensive report detailing the solution for the Complex Computing Problem (CCP) assigned in the CSC-350 & CS-380 Cloud Computing course. This report outlines the design, deployment, and configuration of a personalized cloud-based student portfolio system using Microsoft Azure.

The report is structured in accordance with the assignment requirements, covering Web Application Deployment, Cloud Storage Integration, Secure Networking, and Monitoring and Change Tracking. All necessary evidence, including configuration details and screenshots of the deployed Azure resources, has been included to demonstrate the successful completion of the tasks.

Thank you for your time and consideration.

Sincerely, Junaid Ameer Khan

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2. Part 1: Web Application Deployment
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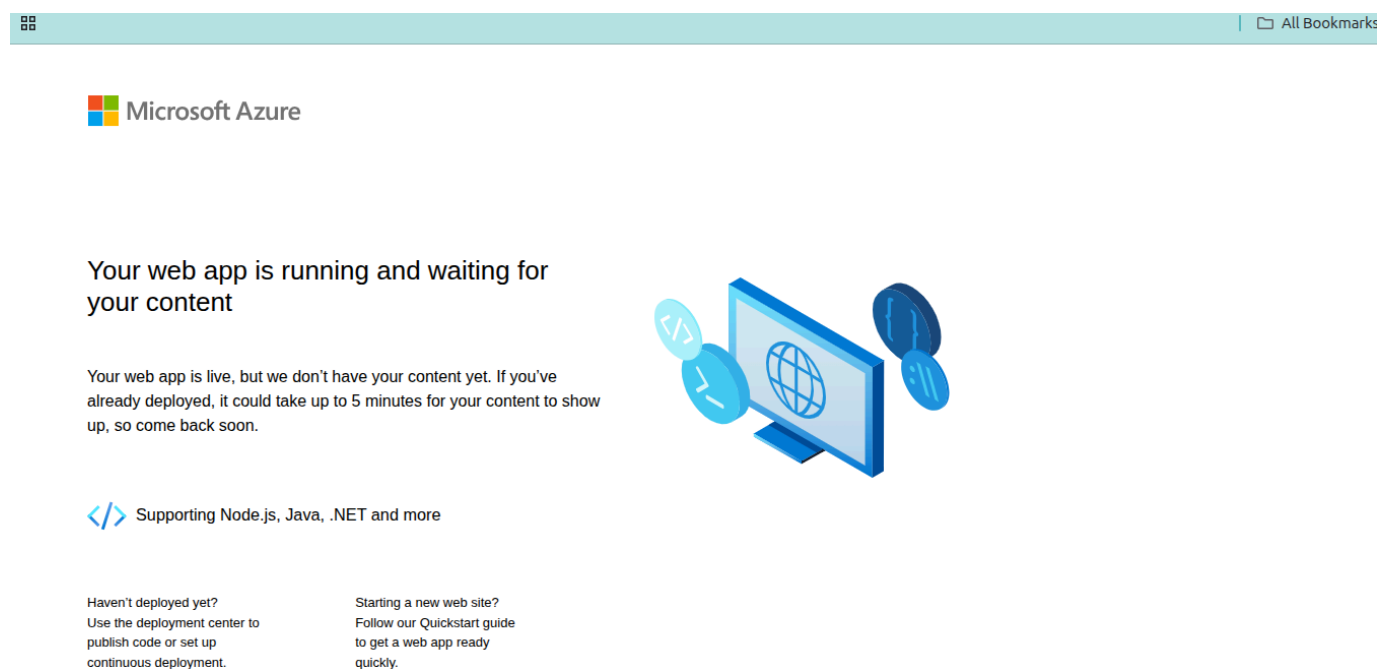
1. Introduction

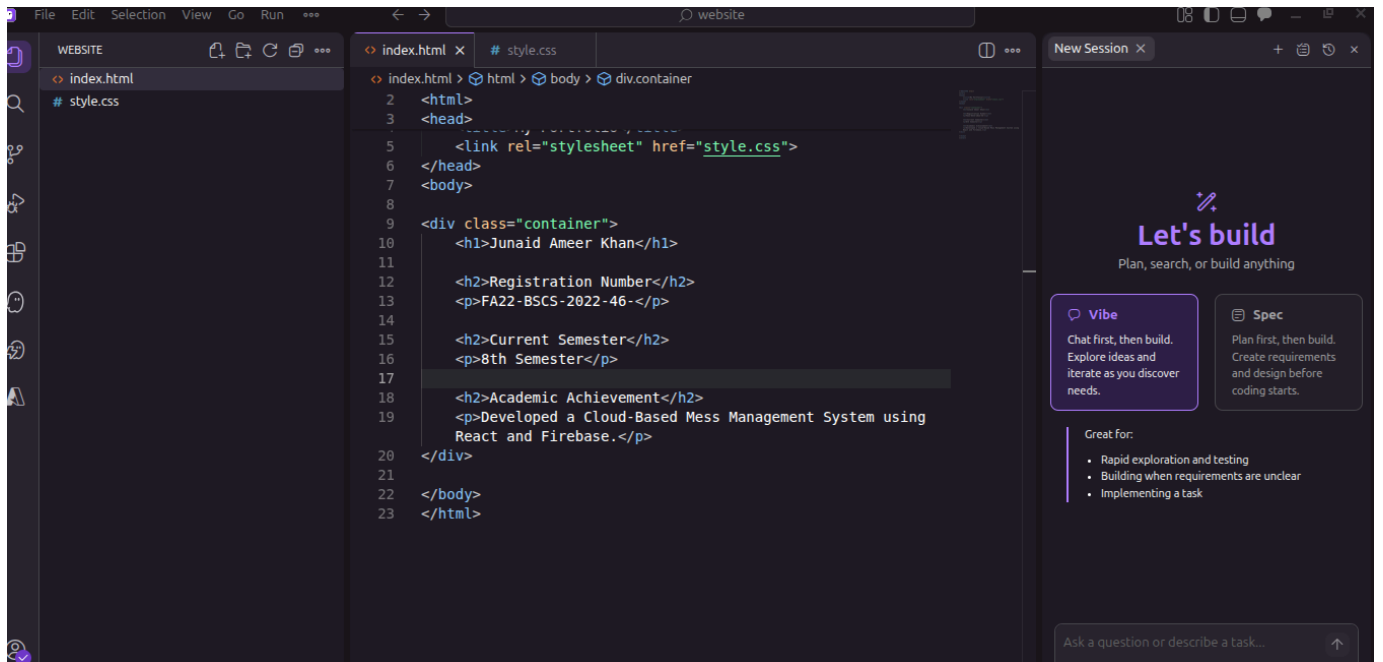
This report presents the implementation details for the cloud-hosted portfolio system designed for Namal University students. The objective of this project is to deploy a personalized portfolio page and implement cloud services to securely store and manage portfolio data using Microsoft Azure. The solution encompasses web hosting, storage configuration, network security, and application monitoring.

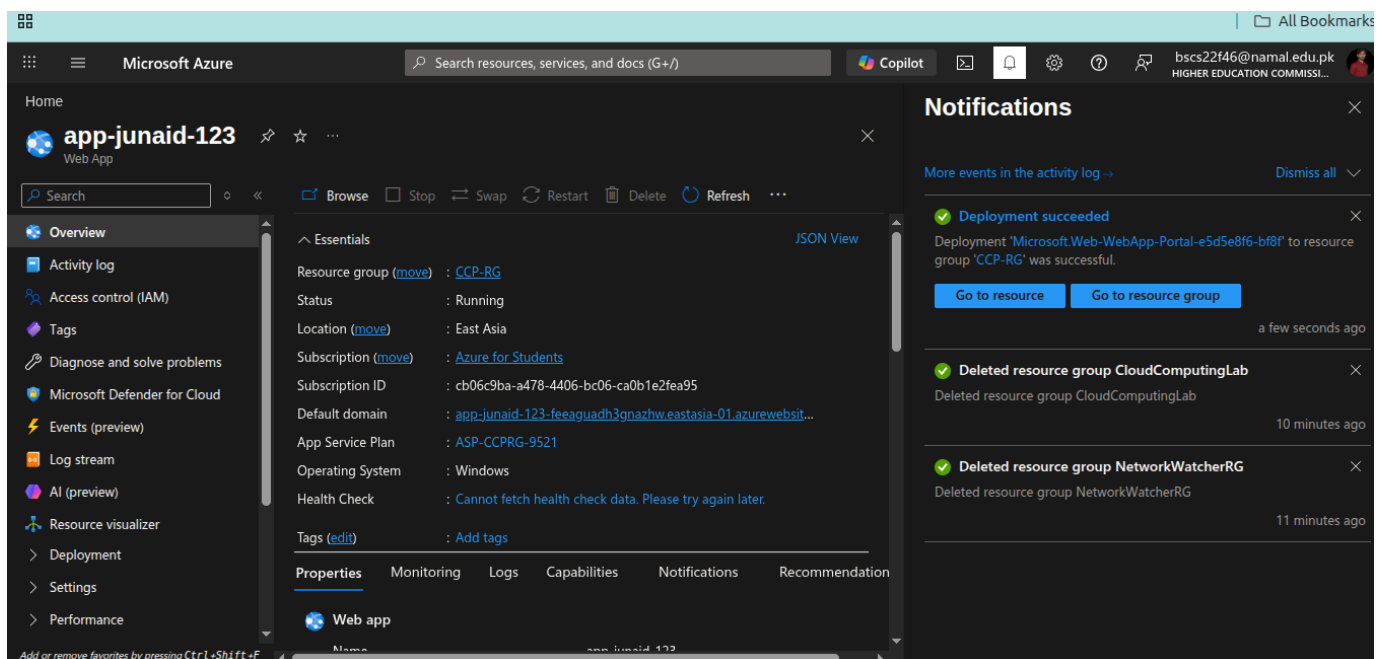
2. Web Application Deployment

The first phase of the project involved developing a personalized web application using HTML and CSS. The application was designed to display the student's name, registration number, current semester, and an academic achievement. The development was carried out using Visual Studio Code.

Following the development, the application was deployed using Azure App Service. The application was named `app-junaaid-123` to adhere to the naming conventions while ensuring global uniqueness. Figure 1 illustrates the Azure App Service placeholder page upon initial deployment. The HTML code used for the personalized portfolio website is shown in Figure 2. The live deployed personalized portfolio website is presented in Figure 3 and Figure 4, showing the initial deployment and after content update, respectively. Figure 5 confirms the successful deployment of the Azure App Service.







3. Cloud Storage Integration

To manage the portfolio data securely, an Azure Storage Account named `storagejunaid46` was created. Within this account, a container named `profile` was established to hold the required text file. A file named `profile.txt`, containing the student's name, registration number, and the date and time of submission, was successfully uploaded.

The replication strategy selected for this storage account was Locally Redundant Storage (LRS). LRS was chosen because it stores three copies of the data within a single datacenter. This strategy ensures durability against hardware failures while keeping costs low. It is highly suitable for academic and development purposes where geo-redundancy is not a strict requirement. Figure 6 shows the Azure Storage Account overview with LRS replication. Figure 7 displays the list of containers within the storage account, including 'profile'. Figure 8 confirms the successful upload of 'profile.txt' to the 'profile' container.

The screenshot displays the Microsoft Azure portal interface for the storage account 'storagejunaaid46'. The left sidebar shows the navigation menu with options like Overview, Activity log, Tags, Diagnose and solve problems, Access Control (IAM), Data migration, Events, Storage browser, Storage Mover, Partner solutions, Resource visualizer, Favorites, and Tables. The main content area shows the 'Essentials' section with various properties and settings.

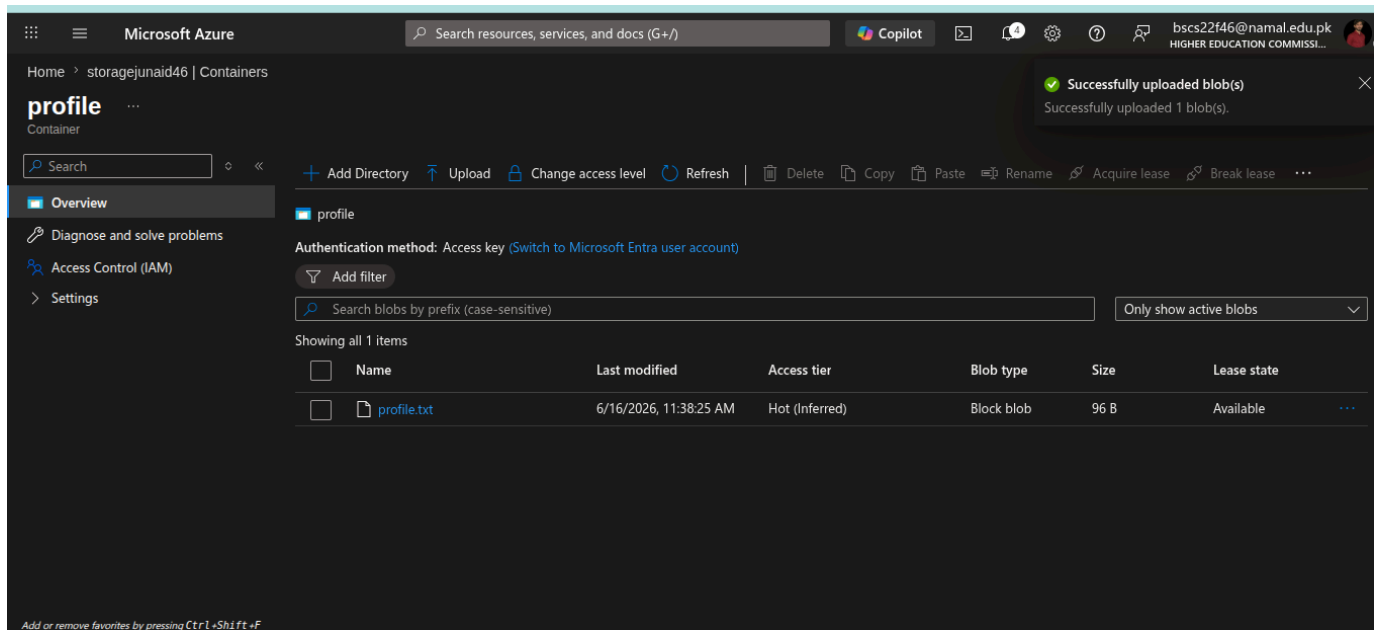
Property	Value
Resource group	CCP-RG
Location	eastasia
Subscription	Azure for Students
Subscription ID	cb06c9ba-a478-4406-bc06-ca0b1e2fea95
Disk state	Available
Tags	Add tags
Performance	Standard
Replication	Locally redundant storage (LRS)
Account kind	StorageV2 (general purpose v2)
Provisioning state	Succeeded
Created	6/16/2026, 11:25:33 AM

The 'Properties' section is expanded, showing the 'Blob service' and 'Security' settings.

Property	Value
Hierarchical namespace	Disabled
Default access tier	Hot
Blob anonymous access	Disabled
Blob soft delete	Disabled
Container soft delete	Disabled
Require secure transfer for REST API operations	Enabled
Storage account key access	Enabled
Minimum TLS version	Version 1.2
Infrastructure encryption	Disabled

The screenshot displays the Microsoft Azure portal interface for the storage account 'storagejunaaid46', specifically the 'Containers' view. The left sidebar shows the navigation menu with options like Overview, Access Control (IAM), Data storage, Containers, Security + networking, Access keys, Settings, Configuration, Resource sharing (CORS), Advisor recommendations, Endpoints, and Help. The main content area shows a list of containers.

Name	Last modified	Anonymous access level	Lease state	Storage connector
\$logs	6/16/2026, 11:25:57 AM	Private	Available	-
profile	6/16/2026, 11:36:05 AM	Private	Available	-



4. Secure Networking

Security is a paramount concern in cloud deployments. To address this, a Virtual Network (VNet) named `VNet-46` was provisioned. Within this VNet, a specific subnet named `WebSubnet` was created to isolate the web application traffic.

To control the inbound and outbound traffic, a Network Security Group (NSG) named `NSG-WebSubnet` was configured and associated with the `WebSubnet`. The NSG was configured with specific inbound security rules:

- Allow HTTP (Port 80): To permit standard web traffic.
- Allow HTTPS (Port 443): To permit secure web traffic.
- Deny All Other Inbound Traffic: To block any unauthorized access attempts.

Figure 9 shows the configuration for adding the `WebSubnet` to the Virtual Network. Figure 10 illustrates the creation of the Network Security Group named `NSG-WebSubnet`. Figure 11 displays the inbound security rules configured for `NSG-WebSubnet`. Figure 12 provides an overview of subnets within the Virtual Network, showing `WebSubnet`. Figure 13 demonstrates associating `NSG-WebSubnet` with the `WebSubnet`. Figure 14 and Figure 15 confirm the successful deployment of the Virtual Network and Network Security Group, respectively.

Microsoft Azure

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HIGHER EDUCATION COMMISSI...

Home > Network foundation | Virtu

Create virtual network

BasicsSecurityAddress space

Virtual network address space

Define the address space of your virtu
segment the virtual network address
into a subnet, Azure assigns the resou

☐ Allocate using IP address pools

+ Add a subnet

10.0.0.0/16

Starting address

10.0.0.0

Select an address space and configure your subnet. You can customize a default subnet or select from subnet templates if you plan to add select services later. [Learn more](#)

Subnet purpose ⓘ

Default

Name ⓘ

WebSubnet

IPv4

Include an IPv4 address space

☒

IPv4 address range ⓘ

10.0.0.0/16

10.0.0.0 - 10.0.255.255

Starting address * ⓘ

10.0.1.0

Size ⓘ

/24 (256 addresses)

Subnet address range ⓘ

10.0.1.0 - 10.0.1.255

IPv6

Include an IPv6 address space

☐ This virtual network has no IPv6 address ranges.

Previous

Next

Add

Cancel

Give feedback

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Home > Network foundation | Network security groups

Create network security group ...

BasicsTagsReview + create

Project details

Subscription *

Azure for Students

Resource group *

CCP-RG

Create new

Instance details

Name *

NSG-WebSubnet

Region *

East Asia

Review + create

< Previous

Next: Tags >

Download a template for automation

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Home > CreateNetworkSecurityGroupBladeV2-20260616115220 | Overview > NSG-WebSubnet

NSG-WebSubnet | Inbound security rules

Network security group

Search

+ Add Hide default rules Refresh Delete Give feedback

Overview

Activity log

Access control (IAM)

Tags

Diagnose and solve problems

Resource visualizer

Settings

Inbound security rules

Outbound security rules

Network interfaces

Subnets

Properties

Locks

Network security group security rules are evaluated by priority using the combination of source, source port, destination, destination port, and protocol to allow or deny the traffic. A security rule can't have the same priority and direction as an existing rule. You can't delete default security rules, but you can override them with rules that have a higher priority.
[Learn more](#)

Filter by name

Port == all Protocol == all Source == all Destination == all Action == all

Priority	Name	Port	Protocol	Source	Destination	Action
100	Allow-HTTP	80	TCP	Any	Any	Allow
110	Allow-HTTPS	443	TCP	Any	Any	Allow
200	Deny-All-Inbound	8080	Any	Any	Any	Deny
65000	AllowVnetInBound	Any	Any	VirtualNetwork	VirtualNetwork	Allow
65001	AllowAzureLoadBalan...	Any	Any	AzureLoadBalancer	Any	Allow
65500	DenyAllInBound	Any	Any	Any	Any	Deny

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Create virtual network

Basics

Security

Address space

Tags

Review + create

10.0.0.0 - 10.0.255.255

65,536 addresses

Subnets	IP address range	Size	NAT gateway
default	10.0.0.0 - 10.0.0.255	/24 (256 addresses)	-
WebSubnet	10.0.1.0 - 10.0.1.255	/24 (256 addresses)	-

Add IPv4 address space

Advertised address prefixes

Previous

Next

Review + create

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Home > Network foundation | Virtual networks > VNet-46

VNet-46 | Subnets

Virtual network

subnet

+ Subnet Refresh

Settings

Connected devices

Subnets

Monitoring

Topology

Create subnets to segment the virtual n
address from the subnet.

Search subnets

☐

Name ↑

☐

default

1

☒

WebSubnet

1

Showing 2 subnets

Save Cancel

Give feedback

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Edit subnet

After March 31, 2026, private subnet will be the default selection for new virtual networks. [Learn more](#)

Security

Simplify internet access for virtual machines by using a network address translation gateway. Filter subnet traffic using a network security group. [Learn more](#)

NAT gateway

A NAT gateway is recommended for outbound internet access from subnets. Edit the subnet to add a NAT gateway. [Learn more](#)

Network security group

Route table

Service Endpoints

Create service endpoint policies to allow traffic to specific azure resources from your virtual network over service endpoints. [Learn more](#)

Services

Remove service endpoint

Microsoft Azure

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Home

VNet-46-1781592446603 | Overview

Deployment

Search

Delete Cancel Redeploy Download Refresh

Overview

Inputs

Outputs

Template

✓ Your deployment is complete

Deployment name : VNet-46-1781592446603

Subscription : Azure for Students

Resource group : CCP-RG

Start time : 6/16/2026, 11:47:32 AM

Correlation ID : e14a6fe4-4090-467c-b4d7-e99705f52e...

> Deployment details

> Next steps

Go to resource

Cost management

Get notified to stay within your budget and prevent unexpected charges on your bill. [Set up cost alerts >](#)

Microsoft Defender for Cloud

Secure your apps and infrastructure [Go to Microsoft Defender for Cloud >](#)

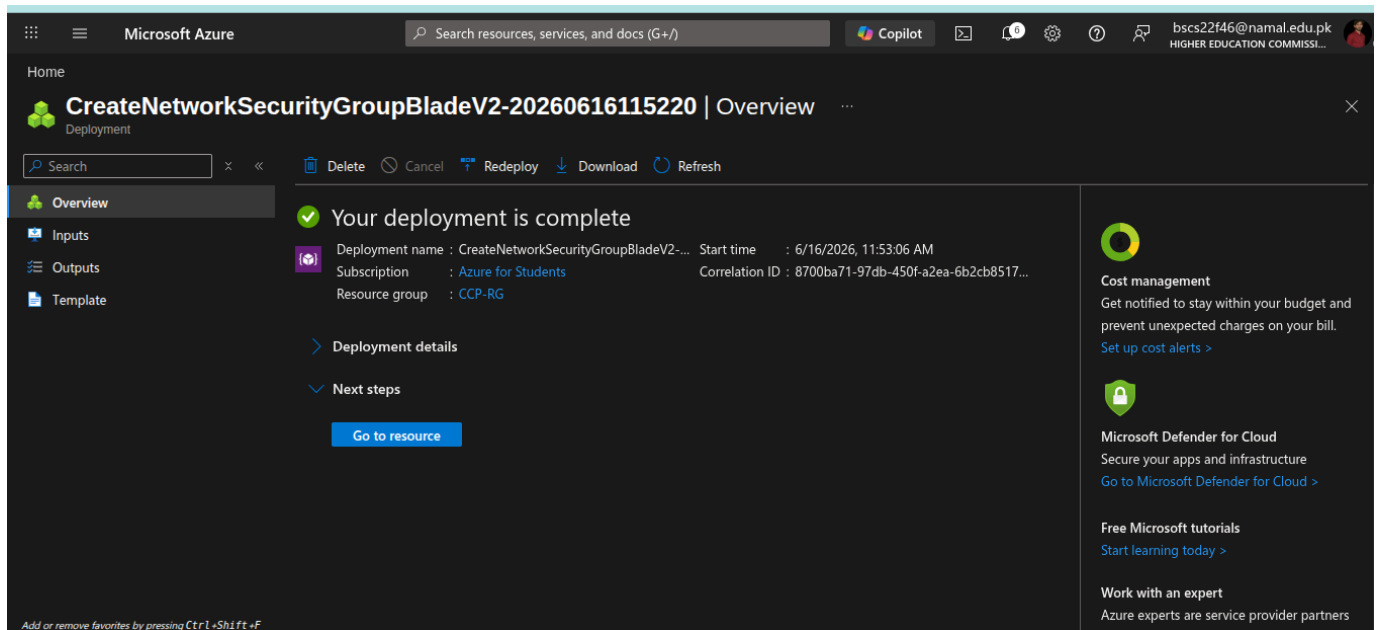
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5. Monitoring and Change Tracking

To ensure the continuous availability and optimal performance of the deployed application, Azure Monitor (Application Insights) was enabled. Monitoring is crucial as it provides real-time insights into application health, helps identify performance bottlenecks, and alerts administrators to potential security threats or downtime.

The primary metric monitored for this application is the Server Response Time. Monitoring this metric is essential to ensure a smooth user experience, as high response times can indicate underlying issues with resource allocation or application code.

Additionally, the portfolio page was updated to include the text “CCP Completed Successfully” to demonstrate the ability to track changes and push updates to the live environment. Figure 16 shows enabling Application Insights for the App Service. Figure 17 displays the Azure Monitor Application Insights dashboard showing key metrics. Figure 18 provides an overview of Application Insights for the deployed web application.

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Home > App Services > app-junaid-123

App Services

Higher Education Commission (perm.onmicrosoft....)

+ Create ▾ ...

☐ Name ↑

☐  app-junaid-123

Showing 1 - 1 of 1. Display count:

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app-junaid-123 | Application Insights

Web App

Search

- > App Service plan
- > Development Tools
- > API
- ▼ Monitoring
 -  Alerts
 -  Metrics
 -  Logs
 -  Health check
 -  Application Insights
 -  Diagnostic settings
 -  App Service logs
 -  Dashboards with Grafana (Preview)
- > Automation

Running final validation...

Application Insights

Collect application monitoring data using Application Insights

Link to an Application Insights resource

 Your app will be connected to an auto-created Application Insights resource: **app-junaid-123**. Instrumentation key will be added to App Settings. This will overwrite any instrumentation key value in web app configuration files.

 As part of using Application Insights instrumentation, we collect and send diagnostic data to Microsoft. This data helps us run and improve Application Insights. You have the option to disable non-essential data collection. [Learn more](#)

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Home > App Services

app-junaid-123

Application Insights

Configure alerts based on specific conditions or thresholds in your application performance data.

Search

★ Application Dashboard  Getting started  Search  Logs  Monitor resource group  Feedback  Favorites  Rename  Delete

30 minutes **1 hour** 6 hours 12 hours 1 day 3 days 7 days 30 days

Overview

- Activity log
- Access control (IAM)
- Tags
- Diagnose and solve problems
- Resource visualizer
- Investigate
- Monitoring
- Usage
- Configure
- Settings
- Automation
- Help

Failed requests

100
80
60
40
20
0

5:30 PM 5:45 PM 6 PM UTC+05:00

Failed requests (Count), app-junaid-123 | 0

Server response time

100ms
80ms
60ms
40ms
20ms
0ms

5:30 PM 5:45 PM 6 PM UTC+05:00

Server response time (Avg), app-jun-123 | --

Server requests

100
80
60
40
20
0

5:30 PM 5:45 PM 6 PM UTC+05:00

Server requests (Count), app-junaid-123 | 0

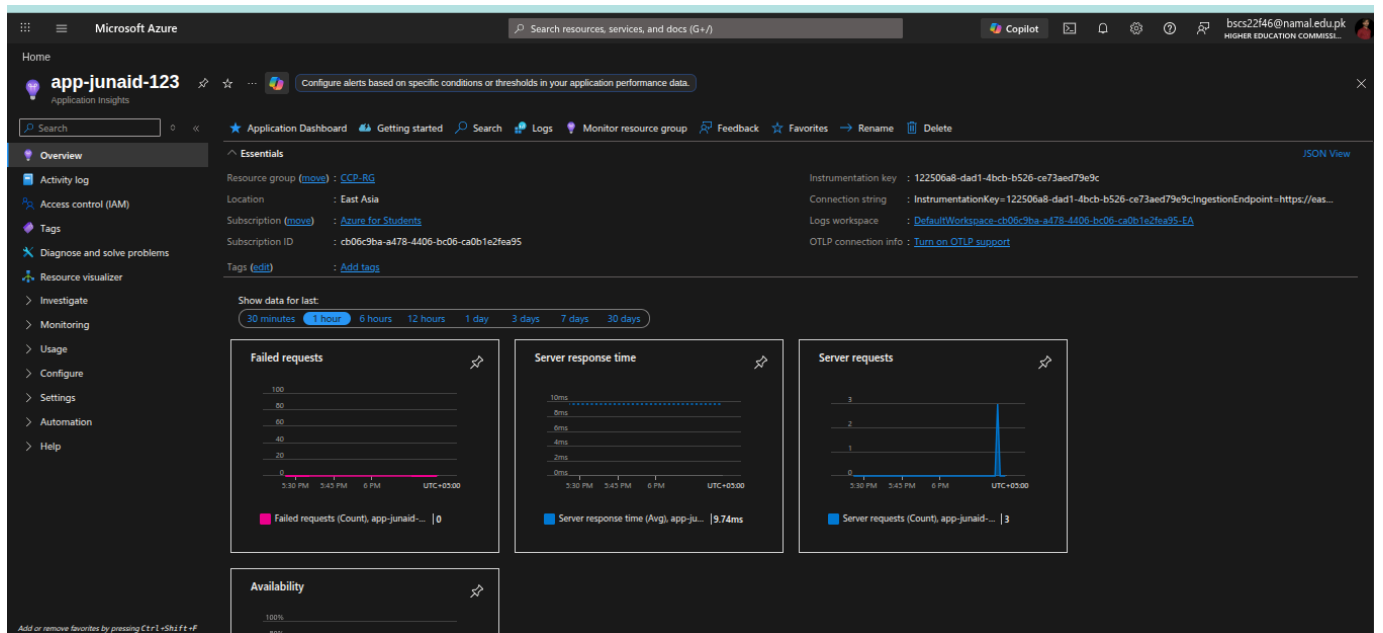
Availability

100%
80%
60%
40%
20%
0%

5:30 PM 5:45 PM 6 PM UTC+05:00

Availability (Avg), app-junaid-123 | --

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6. Conclusion

The deployment of the personalized cloud-based student portfolio system was completed successfully. By leveraging Azure App Service, Azure Storage, Virtual Networks, and Application Insights, a secure, scalable, and monitored environment was established. This project successfully demonstrates the practical application of cloud computing principles in a real-world scenario.